

AI-Based Detection of Earth-Like Exoplanets in Kepler Data

Problem Statement

The search for Earth-like exoplanets using transit photometry is fundamentally limited by the small size of the transit signal and the presence of significant stellar and instrumental noise. For an Earth-sized planet orbiting a Sun-like star at an approximately one-year orbital period, the expected decrease in brightness is extremely small, while the long orbital period means that only a few transits may be observed even in the extensive Kepler dataset. Stellar activity such as sunspots, stellar variability, and instrumental systematics can further obscure or mimic these shallow transit signatures.

This makes the detection of true Earth analogs particularly challenging using conventional transit-search techniques.

Can we do better using modern data-analysis and AI/ML techniques?

The Challenge

You are given real Kepler photometry for 445 stars, split into training, development and evaluation sets. Your task is to build a pipeline that finds transiting planets in that data and characterises what it finds.

You are encouraged to investigate whether improved noise modelling, signal processing, machine learning, deep learning, or other computational approaches can recover weak planetary signals that are difficult to identify using standard methods.

Key challenges include:

- **Very low signal-to-noise ratio.** Earth-sized planets produce extremely shallow transit dips — well below the per-cadence scatter of the light curve.
- **Long transit intervals.** Long orbital periods may result in only a few observable transits across the mission baseline.

- **Stellar variability.** Starspots, flares, rotational modulation and other stellar activity can produce signals that resemble or obscure transits.
- **Instrumental and systematic noise.** Quarter-to-quarter discontinuities, drifts and observational artifacts complicate the identification of genuine periodic signals.
- **Limited number of transits.** Conventional periodicity-based detection becomes difficult when only 3–4 transits are visible.
- **Computational cost.** A full-resolution period search across the complete range is prohibitively expensive. See *Computational Note* below.

The Data

All photometry is **raw SAP flux** from the Kepler mission. It has *not* been detrended, normalised or systematics-corrected. Removing instrumental and stellar signals without destroying the transit is part of the problem, not a preprocessing step that has been done for you.

Each star is a single Parquet file with roughly 65,000 cadences spanning about four years, sampled every 29.4 minutes.

column	meaning
time	days
flux	raw SAP flux
flux_err	uncertainty per measurement
quality	mission quality bitmask; 0 means good
quarter	observing quarter identifier

Released at hour 0

train_pack.zip — 269 stars, with `train_labels.csv` and `train_truth.csv` .

Use this to develop and train.

dev_pack.zip — 89 stars, with `dev_labels.csv` and `dev_truth.csv` .

Use this to evaluate yourself and tune thresholds.

Both packs include full ground truth: which stars host a transit signal, and for every such signal the true period, epoch, depth, duration, planet-to-star radius ratio and number of observed transits.

Released at hour 40

private_pack.zip — 87 stars, identified only as **STAR_0000** through **STAR_0086**. No labels, no truth. Your submission is scored against this set.

The evaluation set contains a mixture of transit signals spanning a wide range of depths, from clearly visible down to near the noise floor, and periods from a few days to several hundred. It also contains stars with no transit signal at all. The proportions are not disclosed.

Scoring

Detection — over all 87 stars.

- Precision, recall and F1 from **prediction**
- PR-AUC and average precision from **confidence**

The **confidence** ranking carries weight comparable to the hard **prediction**. A pipeline that is confident about obvious transits and appropriately uncertain about marginal ones will outrank one that flags the same stars with flat confidence.

Characterisation — over correctly detected stars with known signal parameters.

- Period accuracy: correct if within 2% of the true period or of a $2\times$, $\frac{1}{2}\times$, $3\times$ or $\times$ alias
- Depth accuracy: relative error against true depth

Difficulty breakdown — scores are reported separately across bands of transit depth. Recovering shallow signals is worth substantially more than recovering obvious ones.

Computational Note — read this

A full-resolution Box Least Squares search to 400 days requires on the order of **14 million trial periods**, roughly five hours per star on a typical machine. This is not feasible for any team.

Worse, the obvious workaround fails silently. If you use a coarse fixed grid, your trial periods land too far from the true period, the folded transit smears across phase, and you detect **nothing at all** — even for signals that are

obvious once folded correctly. A pipeline in this state looks identical to one searching empty data.

The required period resolution scales roughly as $P^2 / (\text{baseline} \times \text{duration})$, so cost grows steeply with maximum period. Practical approaches include coarse-to-fine search (a cheap wide sweep to localise candidates, then fine sweeps around each peak) and restricting the searched range.

The starter notebook implements a working coarse-to-fine search. Understand it before modifying it.

Whenever your pipeline reports nothing, verify it against a known injected signal from the training set before concluding the data is empty.

What You Can Do

You are free to design your own approach. Possible directions include:

1. **Noise modelling and removal.** Characterise and suppress stellar and instrumental variability without removing genuine transit signals. This is where the largest gains are available.
2. **AI/ML-based transit detection.** Train models to distinguish planetary transits from stellar variability and noise.
3. **Weak-signal detection.** Identify extremely shallow or incomplete transit signatures that conventional pipelines miss.
4. **Candidate vetting.** Reject false positives using odd/even depth consistency, secondary eclipse searches, per-quarter recurrence and known systematic periods.
5. **Confidence calibration.** Produce probabilities that mean something, since ranking is directly scored.

You do not have to use AI. A statistically or computationally superior non-ML approach is equally valid and will be judged on the same terms.

Target Planet Population

Broad search — rocky, Earth-size to super-Earth. Planets with sizes comparable to Earth, including larger super-Earths, around a broad range of stars. This is the primary scored task.

Focused search — Earth analogs. Earth-sized planets on long orbital periods. Substantially harder — shallow depths combined with few observed transits — but

present in the evaluation set and weighted accordingly in the difficulty breakdown.

You may target either population, or build a pipeline that searches across both.

Expected Outcome

A reproducible pipeline that runs:

```
raw light curve
-> noise / stellar variability modelling
-> signal processing
-> transit detection
-> candidate classification
-> ranked planetary candidates
```

Your submission must report, for each detected candidate: transit depth, duration, orbital period and a detection confidence. Planet-to-star radius ratio follows from depth, and number of transits follows from period and baseline — include them in your write-up where relevant.

Judging

Leaderboard score is one input among several. Submissions are also assessed on methodological soundness, reproducibility and presentation. Weights will be published before the event begins.

Submit a repository with pinned dependencies and a documented entry point. Submissions that cannot be reproduced are ineligible for prizes, regardless of score.

The Ultimate Challenge

The most exciting outcome is an independent planetary candidate not already present in the known candidate or confirmed-planet catalogues.

Can you find a planet hidden in the Kepler data that existing searches have missed?

This is a **separate, optional award**, judged by humans rather than by the scoring script. It requires going beyond the provided dataset to the public

archives.

A claimed candidate must be accompanied by evidence:

- Odd/even transit depth consistency
- Absence of a significant secondary eclipse
- Centroid-shift analysis
- Per-quarter recurrence of the signal
- Cross-match against KOI, TOI, confirmed-planet and eclipsing-binary catalogues
- Contamination check against nearby sources

The organisers make no representation regarding publication or scientific credit for any candidate identified during this event.